

Unbiased Aggregation under Partial Client Participation

Companion proof for: [pleyva2004/communication-efficient-learning-of-deep-networks](https://arxiv.org/abs/2004.08347)

Improvement slug: `unbiased-aggregation`

Abstract

FedAvg’s erratum-corrected server update averages the selected clients’ models with the self-normalized weights n_k/m_t , where $m_t = \sum_{k \in S} n_k$ is a *random* normalizer that depends on which clients were drawn. This is a Hájek ratio estimator of the size-weighted population mean $\bar{v} = \sum_k (n_k/n) v_k$, and a ratio of expectations is not the expectation of a ratio. We prove two complementary facts. (i) The Horvitz–Thompson aggregate $g_{\text{HT}}(S) = (K/m) \sum_{k \in S} (n_k/n) v_k$, which divides each sampled term by its fixed inclusion probability rather than by the random m_t , is *exactly* unbiased for $\bar{v}$ for every population $\{n_k\}$ and every set of client vectors $\{v_k\}$, with no IID assumption. (ii) The self-normalized FedAvg aggregate is exactly unbiased in the balanced regime $n_k \equiv n/K$ (where it collapses to a plain mean), but in general it is biased; we exhibit a two-client counterexample with $m = 1$ that produces a strictly nonzero bias, and explain why the bias is only $O(1/m)$. The take-away is that exact unbiasedness under client imbalance is recovered either by inverse-probability weighting (at the cost of larger per-term weights and hence variance) or, more cheaply, by sampling clients with probability proportional to their size.

Setup and notation

We work entirely at the level of a single communication round of FedAvg, treating the per-client quantities to be aggregated as fixed vectors and the client-selection set S as the only source of randomness. There are K clients indexed $k \in [K] := \{1, \dots, K\}$; client k holds $n_k \geq 1$ examples, and $n := \sum_{k=1}^K n_k$ is the total example count. Associated with client k is a fixed vector $v_k \in \mathbb{R}^d$ — in the FedSGD reading $v_k = \nabla F_k(w_t)$ is the local gradient, and in the FedAvg reading $v_k = w_{t+1}^k$ is the locally updated model; the arguments below are identical in either case because they use only linearity and the partition structure, never any property of v_k itself. The object FedAvg is trying to reconstruct from a subset of clients is the *full size-weighted mean*

$$\bar{v} := \sum_{k=1}^K \frac{n_k}{n} v_k,$$

which (by the exact partition identity $f = \sum_k (n_k/n) F_k$, see the math deep dive §1) is the correct global target: in gradient form $\bar{v} = \nabla f(w_t)$, in model form it is the convex combination that undoes each F_k ’s local normalization. The randomness is the selection set $S \subseteq [K]$. We assume *uniform sampling without replacement of a fixed-size subset*: S is chosen uniformly at random among all $\binom{K}{m}$ subsets of $[K]$ of cardinality exactly m , where $1 \leq m \leq K$ is the (deterministic) number of clients selected per round, $m = \max(CK, 1)$ for participation fraction C . We write $\mathbf{1}\{k \in S\}$ for the inclusion indicator of client k and $\mathbb{E}_S$ for expectation over the draw of S .

Definition 1 (Working notation). For a uniform m -subset S of $[K]$ we use the following.

- **Inclusion probability.** $p_k := \Pr[k \in S] = \mathbb{E}_S[\mathbf{1}\{k \in S\}]$. By symmetry of uniform-without-replacement sampling, p_k is the same for every k . Since $\sum_{k=1}^K \mathbf{1}\{k \in S\} = |S| = m$ for every realization, taking expectations gives $\sum_{k=1}^K p_k = m$, and equal p_k forces $p_k = m/K$ for all k . (Concretely, $p_k = \binom{K-1}{m-1} / \binom{K}{m} = m/K$.)
- **Random subtotal.** $m_t := \sum_{k \in S} n_k = \sum_{k=1}^K \mathbf{1}\{k \in S\} n_k$, the total number of examples held by the selected clients. This is a random variable (it depends on S); its mean is $\mathbb{E}_S[m_t] = \sum_k p_k n_k = (m/K) \sum_k n_k = (m/K) n$.
- **Horvitz–Thompson aggregate.** $g_{\text{HT}}(S) := \frac{K}{m} \sum_{k \in S} \frac{n_k}{n} v_k$, the inverse-probability ($1/p_k = K/m$) reweighting of the sampled size-weighted terms.
- **Self-normalized (FedAvg) aggregate.** $g_{\text{SN}}(S) := \sum_{k \in S} \frac{n_k}{m_t} v_k$, the erratum-corrected server update of FedAvg, whose weights n_k/m_t sum to 1 on every realization but use the *random* normalizer m_t .
- **Balanced regime.** The population $\{n_k\}$ is *balanced* if $n_k \equiv n/K$ for all k (so $n/K \in \mathbb{Z}_{\geq 1}$); otherwise it is *imbalanced*.

Theorem

Theorem 2 (Exact unbiasedness of Horvitz–Thompson; conditional unbiasedness and bias of self-normalized aggregation). *Let S be a uniform m -subset of $[K]$ ($1 \leq m \leq K$) with inclusion probability $p_k = m/K$, and let $v_1, \dots, v_K \in \mathbb{R}^d$ be arbitrary fixed vectors. Then:*

- (i) **(Horvitz–Thompson is exactly unbiased.)** *For every population $\{n_k\}_{k=1}^K$ with $n_k \geq 1$ and every $\{v_k\}$,*

$$\mathbb{E}_S[g_{\text{HT}}(S)] = \bar{v} = \sum_{k=1}^K \frac{n_k}{n} v_k.$$

- (ii) **(Self-normalized: unbiased iff balanced, biased in general.)** *In the balanced regime $n_k \equiv n/K$,*

$$g_{\text{SN}}(S) = \frac{1}{m} \sum_{k \in S} v_k \quad \text{and} \quad \mathbb{E}_S[g_{\text{SN}}(S)] = \bar{v} = \frac{1}{K} \sum_{k=1}^K v_k,$$

i.e. it is exactly unbiased. In general (imbalanced $\{n_k\}$) it is the Hájek ratio estimator $\mathbb{E}_S[A/m_t]$ with $A := \sum_{k \in S} n_k v_k$, and

$$\mathbb{E}_S[g_{\text{SN}}(S)] \neq \bar{v} \quad \text{in general, because} \quad \mathbb{E}_S\left[\frac{A}{m_t}\right] \neq \frac{\mathbb{E}_S[A]}{\mathbb{E}_S[m_t]}.$$

Concretely, for $K = 2$, $m = 1$, sizes $(n_1, n_2) = (1, 3)$ and scalars $(v_1, v_2) = (1, 0)$, the self-normalized estimator has bias $\mathbb{E}_S[g_{\text{SN}}] - \bar{v} = \frac{1}{4} \neq 0$, whereas the Horvitz–Thompson estimator on the same instance is exactly unbiased. The self-normalized bias is in general $O(1/m)$ and vanishes as the number of sampled clients grows.

Proof

Proof. Throughout, the only randomness is the set S ; each n_k , v_k , n , K , and m is a fixed (non-random) constant. We use repeatedly the indicator representation of a sum over the sampled set: for any constants $a_1, \dots, a_K$,

$$\sum_{k \in S} a_k = \sum_{k=1}^K \mathbf{1}\{k \in S\} a_k, \quad (*)$$

which holds because $\mathbf{1}\{k \in S\}$ is 1 exactly for the indices in S and 0 otherwise. We will also use the linearity of expectation over the finite index set $[K]$ and the basic fact $\mathbb{E}_S[\mathbf{1}\{k \in S\}] = \Pr[k \in S] = p_k = m/K$ established in the Working notation.

Part (i): the Horvitz–Thompson aggregate is exactly unbiased.

Start from the definition and rewrite the sum over S as a sum over all of $[K]$ using $(*)$ with $a_k = (n_k/n)v_k$, then pull the deterministic factor K/m inside:

$$g_{\text{HT}}(S) = \frac{K}{m} \sum_{k \in S} \frac{n_k}{n} v_k = \frac{K}{m} \sum_{k=1}^K \mathbf{1}\{k \in S\} \frac{n_k}{n} v_k = \sum_{k=1}^K \frac{K}{m} \mathbf{1}\{k \in S\} \frac{n_k}{n} v_k.$$

Take the expectation $\mathbb{E}_S$. The vector $v_k \in \mathbb{R}^d$ is taken coordinatewise; expectation is linear on each coordinate and hence on the finite sum of vectors, and the only random factor in the k -th summand is the scalar indicator $\mathbf{1}\{k \in S\}$ (everything else — K , m , n_k , n , v_k — is constant). Therefore

$$\mathbb{E}_S[g_{\text{HT}}(S)] = \sum_{k=1}^K \frac{K}{m} \frac{n_k}{n} v_k \mathbb{E}_S[\mathbf{1}\{k \in S\}] = \sum_{k=1}^K \frac{K}{m} \frac{n_k}{n} v_k \cdot p_k.$$

Now substitute $p_k = m/K$. The inverse-probability weight K/m and the inclusion probability m/K cancel *exactly* for every k simultaneously:

$$\frac{K}{m} \cdot p_k = \frac{K}{m} \cdot \frac{m}{K} = 1.$$

Hence

$$\mathbb{E}_S[g_{\text{HT}}(S)] = \sum_{k=1}^K \frac{n_k}{n} v_k = \bar{v}.$$

No property of $\{n_k\}$ beyond $n_k \geq 1$ (so that $n > 0$ and the weights are well defined) and no property of $\{v_k\}$ was used; in particular no IID assumption on the data partition enters. The cancellation is purely the design of the inverse-probability weight against the (deterministic, equal) inclusion probability, so unbiasedness holds for an arbitrary, even adversarial, population. This is the classical Horvitz–Thompson estimator from survey sampling (Horvitz and Thompson 1952), here in its degenerate form with a deterministic normalizer; the federated framing matches the unbiased partial-participation gradient discussed by, e.g., Karimireddy et al. (2020). This proves (i).

Part (ii), balanced case: the self-normalized aggregate reduces to a plain mean and is exactly unbiased.

Suppose $n_k \equiv c$ for the common value $c := n/K \in \mathbb{Z}_{\geq 1}$. Then the random subtotal collapses to a constant on every realization, because S has exactly m elements:

$$m_t = \sum_{k \in S} n_k = \sum_{k \in S} c = c|S| = cm,$$

which no longer depends on *which* clients are in S . The self-normalized weight of each selected client is therefore the constant

$$\frac{n_k}{m_t} = \frac{c}{cm} = \frac{1}{m},$$

so the self-normalized aggregate is the plain (unweighted) mean over the selected clients,

$$g_{\text{SN}}(S) = \sum_{k \in S} \frac{n_k}{m_t} v_k = \frac{1}{m} \sum_{k \in S} v_k.$$

Now take the expectation, again rewriting the sum over S via $(*)$ with $a_k = v_k$ and using linearity and $\mathbb{E}_S[\mathbf{1}\{k \in S\}] = p_k = m/K$:

$$\mathbb{E}_S[g_{\text{SN}}(S)] = \frac{1}{m} \sum_{k=1}^K v_k \mathbb{E}_S[\mathbf{1}\{k \in S\}] = \frac{1}{m} \sum_{k=1}^K v_k \cdot \frac{m}{K} = \frac{1}{K} \sum_{k=1}^K v_k.$$

Finally, in the balanced regime the target itself simplifies: $\bar{v} = \sum_k (n_k/n) v_k = \sum_k (c/(cK)) v_k = \frac{1}{K} \sum_k v_k$. Hence $\mathbb{E}_S[g_{\text{SN}}(S)] = \bar{v}$ exactly. In this regime the self-normalized aggregate and the Horvitz–Thompson aggregate coincide: substituting $n_k \equiv c$ into g_{HT} gives $g_{\text{HT}}(S) = \frac{K}{m} \sum_{k \in S} \frac{c}{cK} v_k = \frac{1}{m} \sum_{k \in S} v_k = g_{\text{SN}}(S)$, so both are the same plain mean. This is exactly the regime of the paper’s MNIST/CIFAR experiments, where $n_k \equiv n/K$ by construction, which is why FedAvg’s footnote-2 unbiasedness claim is correct there.

Part (ii), general case: the self-normalized aggregate is a ratio estimator and is biased.

For an imbalanced population the normalizer $m_t = \sum_{k \in S} n_k$ genuinely varies with S . Write the (vector) numerator and (scalar) denominator

$$A := \sum_{k \in S} n_k v_k = \sum_{k=1}^K \mathbf{1}\{k \in S\} n_k v_k, \quad m_t = \sum_{k \in S} n_k = \sum_{k=1}^K \mathbf{1}\{k \in S\} n_k,$$

so that $g_{\text{SN}}(S) = A/m_t$ is a ratio of two random quantities driven by the *same* draw S . This is precisely the Hájek ratio estimator of survey sampling (Hájek 1971). The expectations of numerator and denominator are easy: by linearity and $p_k = m/K$,

$$\mathbb{E}_S[A] = \sum_{k=1}^K \frac{m}{K} n_k v_k = \frac{m}{K} \sum_{k=1}^K n_k v_k = \frac{m}{K} n \bar{v}, \quad \mathbb{E}_S[m_t] = \sum_{k=1}^K \frac{m}{K} n_k = \frac{m}{K} n.$$

Their ratio is exactly the target,

$$\frac{\mathbb{E}_S[A]}{\mathbb{E}_S[m_t]} = \frac{(m/K) n \bar{v}}{(m/K) n} = \bar{v}.$$

The bias of g_{SN} is the gap between the *expectation of the ratio* and this *ratio of expectations*:

$$\mathbb{E}_S\left[\frac{A}{m_t}\right] - \bar{v} = \mathbb{E}_S\left[\frac{A}{m_t}\right] - \frac{\mathbb{E}_S[A]}{\mathbb{E}_S[m_t]}.$$

This gap is generically nonzero because $x \mapsto 1/x$ is strictly convex on $(0, \infty)$ and A and m_t are positively dependent (a draw of large clients inflates both A and m_t together), so the expectation of the quotient does not factor as the quotient of expectations. We now make the inequality concrete with an explicit instance, which is the cleanest possible disproof of unbiasedness in general.

Two-client, $m = 1$ counterexample. Take $K = 2$, $m = 1$ (so we sample exactly one of the two clients, each with probability $p_1 = p_2 = m/K = \frac{1}{2}$), sizes $(n_1, n_2) = (1, 3)$ so $n = 4$, and scalar payloads $(v_1, v_2) = (1, 0)$. The true target is

$$\bar{v} = \frac{n_1}{n}v_1 + \frac{n_2}{n}v_2 = \frac{1}{4} \cdot 1 + \frac{3}{4} \cdot 0 = \frac{1}{4}.$$

With $m = 1$ the selected set is either $S = \{1\}$ or $S = \{2\}$, each with probability $\frac{1}{2}$, and in each case the single selected client gets self-normalized weight $n_k/m_t = n_k/n_k = 1$. Hence the self-normalized aggregate is simply the payload of the chosen client:

$$g_{\text{SN}}(\{1\}) = \frac{n_1}{n_1}v_1 = v_1 = 1, \quad g_{\text{SN}}(\{2\}) = \frac{n_2}{n_2}v_2 = v_2 = 0.$$

Therefore

$$\mathbb{E}_S[g_{\text{SN}}] = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 0 = \frac{1}{2}, \quad \text{bias} = \mathbb{E}_S[g_{\text{SN}}] - \bar{v} = \frac{1}{2} - \frac{1}{4} = \frac{1}{4} \neq 0.$$

The mechanism is transparent: self-normalization discards the size information n_k entirely once a client is alone in S (its weight is forced to 1), so the small client 1 and the large client 3 are treated as equals, over-weighting the small client relative to its true share $n_1/n = \frac{1}{4}$. The Horvitz–Thompson aggregate avoids this. On the same instance, $g_{\text{HT}}(\{1\}) = \frac{K}{m} \frac{n_1}{n} v_1 = \frac{2}{1} \cdot \frac{1}{4} \cdot 1 = \frac{1}{2}$ and $g_{\text{HT}}(\{2\}) = \frac{2}{1} \cdot \frac{3}{4} \cdot 0 = 0$, so $\mathbb{E}_S[g_{\text{HT}}] = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot 0 = \frac{1}{4} = \bar{v}$, confirming Part (i) numerically and isolating the bias as a property of the random normalizer, not of the sampling.

Order of the bias. The bias is not merely nonzero; it shrinks like $O(1/m)$ as the number of sampled clients grows, which is why FedAvg’s self-normalized rule is still *consistent*. Write the centered numerator and denominator $A = \mathbb{E}_S[A] + \delta_A$, $m_t = \mathbb{E}_S[m_t] + \delta_m$ with $\mathbb{E}_S[\delta_A] = 0 = \mathbb{E}_S[\delta_m]$, and abbreviate $\mu_A = \mathbb{E}_S[A]$, $\mu_m = \mathbb{E}_S[m_t] > 0$. A first-order Taylor expansion of $1/m_t$ about μ_m gives, for $|\delta_m| < \mu_m$,

$$\frac{1}{m_t} = \frac{1}{\mu_m + \delta_m} = \frac{1}{\mu_m} \left(1 - \frac{\delta_m}{\mu_m} + \frac{\delta_m^2}{\mu_m^2} - \dots \right),$$

so, multiplying by $A = \mu_A + \delta_A$ and taking expectations (using $\mathbb{E}_S[\delta_A] = \mathbb{E}_S[\delta_m] = 0$ to kill the linear terms),

$$\mathbb{E}_S \left[\frac{A}{m_t} \right] = \frac{\mu_A}{\mu_m} - \frac{\text{Cov}_S(A, m_t)}{\mu_m^2} + \frac{\mu_A \text{Var}_S(m_t)}{\mu_m^3} + \dots = \bar{v} + \frac{1}{\mu_m^2} \left(\bar{v} \text{Var}_S(m_t) - \text{Cov}_S(A, m_t) \right) + \dots$$

Under uniform-without-replacement sampling of m clients, A and m_t are each sums of m exchangeable contributions, so $\text{Var}_S(m_t)$ and $\text{Cov}_S(A, m_t)$ scale like m (times a finite-population correction), while $\mu_m^2 = (m n/K)^2$ scales like m^2 ; hence the leading bias term is of order $m/m^2 = 1/m$. Thus the self-normalized FedAvg aggregate has bias $\Theta(1/m)$ in general — exactly zero when balanced (where, as shown, $\text{Var}_S(m_t) = 0$ and $\text{Cov}_S(A, m_t)$ contributes the matching plain-mean value so the bracket vanishes identically) and vanishing as more clients are sampled per round. This completes the proof of (ii) and of the theorem. $\square$

Discussion

What this gives us. We have an exact, assumption-light recipe for unbiased aggregation under partial client participation. Part (i) shows the Horvitz–Thompson aggregate $g_{\text{HT}}(S) = (K/m) \sum_{k \in S} (n_k/n) v_k$

reconstructs the true size-weighted target $\bar{v}$ in expectation for *any* population $\{n_k\}$ and *any* payloads $\{v_k\}$ — gradients or whole models, IID or arbitrarily non-IID partitions — because the cancellation $(K/m) \cdot p_k = 1$ is purely combinatorial. Part (ii) pins down precisely when FedAvg’s shipped rule is safe: it is exactly unbiased in the balanced regime $n_k \equiv n/K$ that the paper’s MNIST/CIFAR experiments use (there it is literally a plain mean over sampled clients), so FedAvg’s footnote-2 unbiasedness claim is correct in exactly the setting the experiments occupy. Two practical fixes follow for the imbalanced case: inverse-probability reweighting (server-side, stateless, changes only the coefficients), or sampling clients with probability proportional to n_k (after which the plain mean is already unbiased per draw). The explicit $m = 1$, two-client counterexample makes the failure tangible and auditable rather than asymptotic.

What this does not give us. Unbiasedness is a statement about the *mean* of the one-round aggregate; it says nothing about variance, and nothing about the multi-round dynamics of FedAvg. In particular: (a) it does not address the nonlinearity of $E > 1$ local steps — here each v_k is treated as a fixed vector, so the result applies verbatim to the gradient ($E = 1$) reading and to the model-averaging reading *conditional on* the local updates, but it does not by itself make the iterated FedAvg map an unbiased estimator of centralized training (that is the separate heterogeneity-drift question, deep dive §3). (b) It gives no convergence rate; unbiased per-round aggregation does not imply FedAvg converges, only that the aggregation step introduces no systematic shift. (c) The $O(1/m)$ bound on the self-normalized bias is a leading-order Taylor characterization with a finite-population scaling argument for the moments; it identifies the order, not a sharp constant, and assumes m_t stays bounded away from 0 (guaranteed here since every $n_k \geq 1$, so $m_t \geq m \geq 1$).

Where it would break. The whole argument hinges on the inclusion probabilities being known and equal to $p_k = m/K$, which requires uniform sampling without replacement of a *fixed-size* set S with $|S| = m$ deterministic. If clients are instead sampled with replacement, or each client participates independently with probability q (so $|S|$ is random), or availability/stragglers make participation data-dependent, then p_k changes (and may even be unknown), and the Horvitz–Thompson weight $1/p_k$ must be recomputed accordingly — using K/m blindly would reintroduce bias. If p_k is genuinely unknown, neither estimator is provably unbiased. The exact-unbiasedness of the self-normalized rule additionally breaks the moment n_k are not all equal, exactly as the counterexample shows; and the $O(1/m)$ control degrades if the size distribution is heavy-tailed enough that a single giant client dominates m_t (then $\text{Var}_S(m_t)/\mu_m^2$ need not shrink like $1/m$). Finally, the Horvitz–Thompson fix trades bias for variance: its per-term weight is $K/(m n) \cdot n_k$, up to a factor $K/m = 1/C$ larger than the self-normalized weight, so at small participation C it inflates variance — which is exactly why size-proportional sampling ($p_k \propto n_k$, after which the plain mean is unbiased without any large $1/p_k$ factor) is the variance-friendly alternative when client sizes are very uneven.

Empirical companion. This proof is the validation file for the improvement unbiased-aggregation in `05-improvements.tex`. If a measurement-mode prototype also exists at `improvements/unbiased-aggregation` it serves as a sanity check, not as evidence — the theorem above is the load-bearing claim.